

Sentiment Analysis Tool: Technical Explanation

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What it is

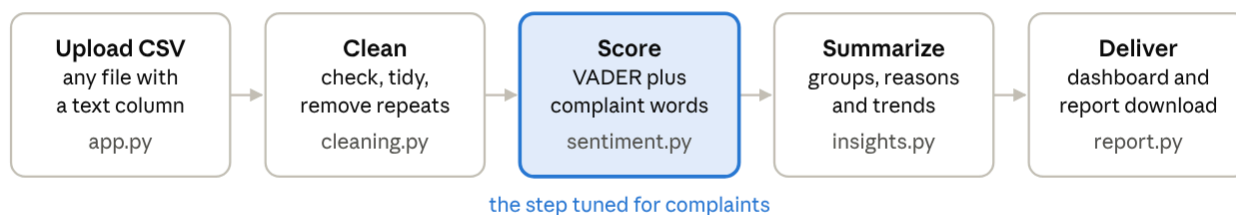
The Sentiment Analysis Tool takes a CSV file of written comments, cleans it, labels every comment as positive, neutral or negative, shows the results on a dashboard, and writes a report the user can download. It works on any CSV with a column of text, such as tweets, reviews or survey answers, and was tested on the public Kaggle "Twitter US Airline Sentiment" file of 14,640 tweets.

It is the Week 3 individual project of the CAPACITI x Clickatell AI Bootcamp, and it exists as both a Streamlit dashboard and a command-line script that share the same code.

- **Live app:** samukelo-sentiment-analysis-tool.streamlit.app
- **Code:** github.com/mngadisamu/sentiment-analysis-tool
- **Built with:** Python, pandas, VADER, Streamlit, Plotly

How it works

Five steps turn a CSV into a dashboard and a report



app.py (dashboard) and analyze.py (command line) run these same steps.

pipeline of the tool · 5 steps and the file behind each

Each step hands its result to the next. The dashboard and the command-line script call the same functions in the same files, so they always give the same numbers. The original file is never changed: cleaning works on a copy, and scoring adds three columns (`clean_text` , `sentiment` and `score`) to that copy.

Data cleaning

Cleaning always works on a copy, so the uploaded file is never changed, and every step is written to a log that shows how many rows it affected. The log appears in the dashboard and in the report, so each number can be explained.

Step	What it does	Why	Result on the Kaggle file
File check	Counts rows, duplicates, blank texts and missing values, and warns if the chosen column does not look like free text	Catches the wrong column or a damaged file before any analysis	14,640 rows, 36 duplicates, no blank texts
Tidy spaces	Removes leading, trailing and double spaces in every text column	Stops "Delta" and "Delta " counting as different values	2,339 cells changed
Standardize spelling	In category columns, maps minority spellings to the most common one, so "late flight" becomes "Late Flight"	Keeps groups and reasons from splitting into near-duplicates	None needed
Remove blank text	Drops rows with no text	There is nothing to score	None
Remove duplicate rows	Drops exact copies, keeping the first	Duplicates count one opinion twice	36 removed
Remove repeated texts	Drops rows with the same text as an earlier row	Optional and off by default, because different people can write the same words	Not used
Read dates	Reads the date column as real dates	Needed for the sentiment over time chart	14,604 dates read

Dates are read as written: a time-zone offset at the end, such as -0800, is ignored. An earlier version converted to UTC, which moved a tweet written at 4:20 pm on 24 Feb onto 25 Feb. Each cleaning step can be switched off in the dashboard.

Scoring sentiment

Each text is scored with VADER, a rule-based method built for social media text. It looks up every word in a dictionary of word scores, adjusts for things like negation, capital letters and exclamation marks, and combines them into one compound score from -1 (very negative) to +1 (very positive).

The tool turns the score into a label using VADER's recommended thresholds: 0.05 or more is positive, -0.05 or less is negative, and anything between is neutral. Before scoring, links, @mentions and hashtag symbols are removed, and leftover web codes such as the one for the & sign are turned into real characters.

I chose a rule-based method over a trained model for three reasons. It needs no training data, so it works on any CSV. Every score can be explained from the words in the text. And it runs fast enough on a free hosting plan.

Customer messages mode

Testing on the Kaggle file showed that plain VADER was too positive on complaints. It does not know that a "cancelled" or "delayed" flight is bad news, and it counts "help" and "please" as positive even when a customer is asking for help with a problem. Customer messages mode, on by default in the dashboard, adds a hand-written list of 36 common complaint words (such as cancelled, delayed, lost, stuck, refund) with negative scores, and sets "help" and "please" to neutral. Words not on the list keep VADER's own score. It can be switched off for text that is not complaints or support messages.

Testing and results

The Kaggle file includes a human-written sentiment label for every tweet, so the tool's labels could be checked against them. This is the "agreement" figure in the dashboard and report, and it measures how often the tool and the dataset give the same label.

The first version, with plain VADER, agreed on 55.0% of 14,604 tweets. It found only 51% of the tweets labelled negative, and rated 35.5% of tweets negative where the labels say about 63%. Changes were tried one at a time and each was re-measured:

Change tried	Agreement	Kept?
Plain VADER	55.0%	Starting point
Turn web codes into characters	55.0%	Yes, for cleaner word counts

Change tried	Agreement	Kept?
Treat "thanks", "help", "please" and similar as neutral	55.9%	No, it lost 20 points of positive tweets (89% found down to 69%)
Add the complaint word list, and "help" and "please" neutral	about 65%	Yes
Change the positive and negative thresholds	0.6 points better at most	No, not worth the extra setting

The final tool agrees on 65.9% and finds 69% of the tweets labelled negative, up from 51%. It rates 49.0% of tweets negative.

Two cautions apply. The word list was written after looking at the most common words in this file, so the figure is measured on data that informed it, and a smaller gain is likely on other data. And the tool still rates fewer tweets negative than the labels do, so it is best used to compare airlines and days rather than to state absolute shares. The insights report has the full results.

Dashboard and report

The dashboard has four tabs: Dashboard, Data quality and cleaning, Report, and Data. The sidebar walks through five choices in order.

1. **Data.** Upload a CSV or use the built-in sample, then choose the text column.
2. **Columns.** Optionally choose the columns to compare groups by, to explain negative texts, to check against existing labels, and to plot over time. The tool guesses these from the column names and contents, and the user can change any guess or choose none. This is what lets it work on any dataset and not only airline tweets.
3. **Cleaning.** Switch each cleaning step on or off.
4. **Filters.** Keep only chosen values of the group column, such as two airlines.
5. **Report.** Set a title and tick which sections to include.

The dashboard charts are the sentiment breakdown, sentiment over time, sentiment by group and the top reasons for negative texts. The user can download the cleaned data, the results with the sentiment columns added, and the report as HTML or Markdown.

The report has up to 11 sections: overview, data quality, cleaning, breakdown, over time, by group, reasons, common words, example texts, accuracy check and method. A section is left out automatically when the file has no column for it, and the report says what was left

out and why. The wording is generated from templates filled in with the numbers, not by an AI model, so the same data always gives the same report. Text from the file is escaped before it goes into the HTML report, so a tweet containing code cannot break the page.

Design decisions and limitations

Design decisions

- **One core, two front ends.** The dashboard and the command-line script call the same cleaning, scoring, insights and report code, so they always give the same results.
- **Nothing is overwritten.** Cleaning works on a copy. If the uploaded file already has columns called `sentiment`, `score` or `clean_text`, they are kept as `original_sentiment` and so on.
- **Every change is logged.** The cleaning log and the method section let a reader see what was done to the data.
- **Columns are guessed, then confirmed.** Guessing saves effort, and the sidebar lets the user correct a wrong guess.
- **Small groups are flagged.** The report adds a caution when a group or period has few texts, because percentages from a handful of texts are unreliable.

Limitations

- VADER scores words, so it misses sarcasm, context and slang, and it works on English only.
- Even with customer messages mode, the tool agrees with the Kaggle labels on 65.9% of tweets and rates fewer tweets negative than the labels do. Use it to compare groups and days, not to state absolute shares.
- The complaint word list was written after looking at this dataset, so expect a smaller gain on other data.
- The tool does not explain why sentiment changes. The data shows a peak in negative tweets on 22 Feb 2015 but not its cause.
- Next steps: test a model trained on tweets against the 65.9%, and check the tool on a second labelled dataset.

Running and deploying it

The live app needs nothing installed. To run it on a computer with Python 3.9 or newer:

```
pip install -r requirements.txt
streamlit run app.py
```

The command-line version runs the same steps without the dashboard:

```
python analyze.py Tweets.csv --report report.html
```

It guesses the columns, and options such as `--group-column`, `--date-column`, `--groups`, `--sections` and `--general-text` change that. The README lists them all.

The app is deployed on Streamlit Community Cloud from the GitHub repository. Pushing a change to the `main` branch redeploys it automatically. The main file is `app.py`, and the packages are listed in `requirements.txt` (pandas, vaderSentiment, markdown, streamlit and plotly). The Kaggle file itself is not in the repository, since it is large and belongs to Kaggle, so a small illustrative sample is included instead.